Storytelling with Multi-Sensory AR

This project combines augmented reality and interactive storytelling to create an immersive, multisensory experience. The project emphasises user participation, accessibility, and collaboration, highlighting the potential of AR in art while exploring human-computer interaction and inclusive design.

Tools:

Keywords:
#Human-Computer Interaction #Augmented Reality #Interactive Storytelling #Accessibility #Multimedia Art #ParticipatoryDesign

Website:

https://canacechen.com/beforethesunrises.html

*This is an individual academic project.

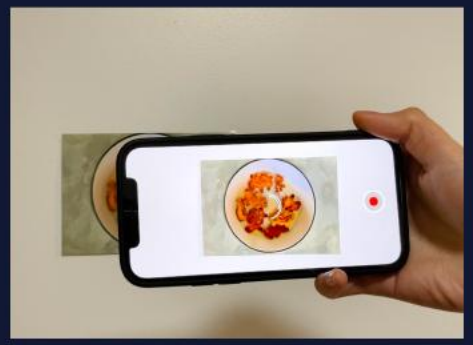
Research

Inspiration

The inspiration for the artwork was apple's live photo. When the user presses and holds a live photo, a short animation from the time when the photo was taken will be played.

Multi-sensory AR as an accessible medium

AR empowers the connection between the human senses and technology; it blends digital information with the real world for people to have a unique perspective. This gives AR the potential to influence people emotionally and socially (Pucihar and Kljun, 2018). Choosing AR allows users to experience the artwork with multiple senses, enabling a more engaging and immersive experience while enhancing inclusiveness for diverse users.



AR for mobile phone user

Since AR is displayed using a mobile device, the screen will be relatively small and the user will have one of the hands occupied. In this case, it would be easier for the user to move the device relative to the real world rather than moving their fingers on the screen (Henrysson, Billinghurst and Ollila, 2005).

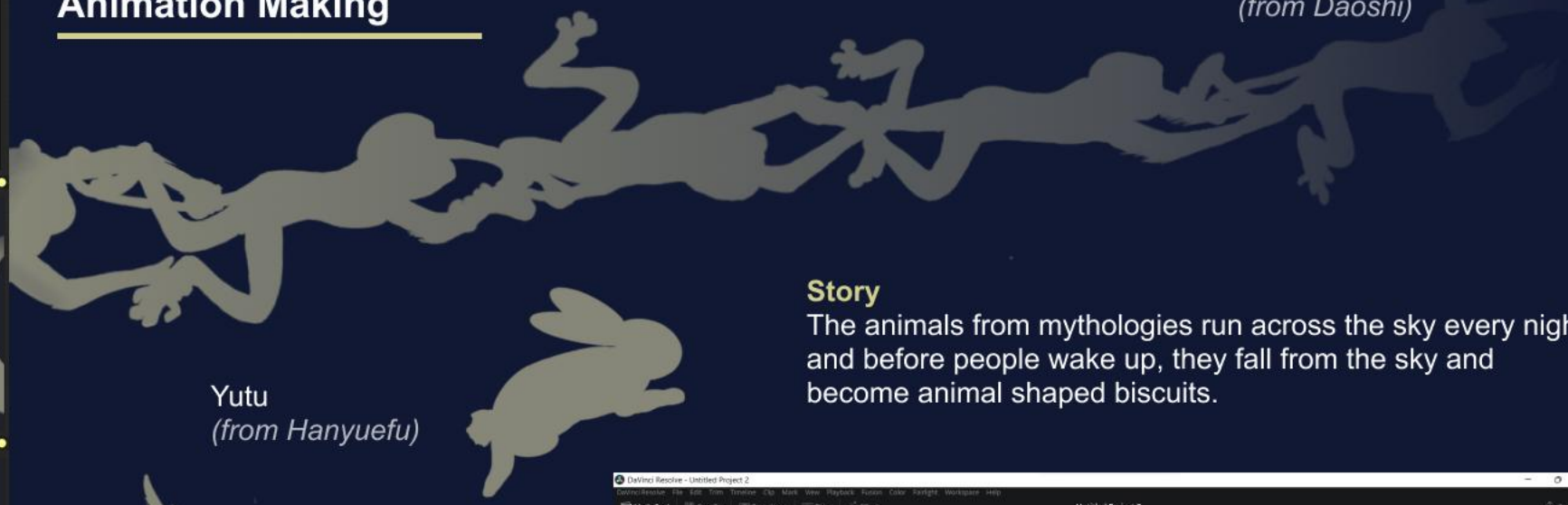
Collaborative AR

The interface needs to display its components with clear visibility and information so that the users will understand the purpose of each component and find the right one to use. The tools should not take too many interactions to use because having faster switches among the them will allow the users to multitask better (Jacucci, 2005).

User Groups

User Group	Needs and Wants
General public	• Engaging narrative and visually appealing experience • Clear and easy-to-follow instructions to ensure understanding • Seamless interaction between physical and digital elements
Art/Technology Enthusiast	• Smooth, intuitive, and logical operation • Enjoyable exploration of interactive art and technology • Ability to understand both the technical and artistic aspects of the experience
Families	• Age-appropriate content that is entertaining and accessible to all family members • Simple and intuitive user interface for ease of interaction • Non-technical instructions for smooth usage by all age groups
Students	• Inspiring and thought-provoking content • A well-organized, clear, and engaging user journey
People with disabilities	• inclusive design • Comprehensive accessibility research to meet diverse needs
Elderly Users	• Larger text and buttons for better readability and ease of navigation • Simple interactions with minimal steps to avoid confusion

Animation Making



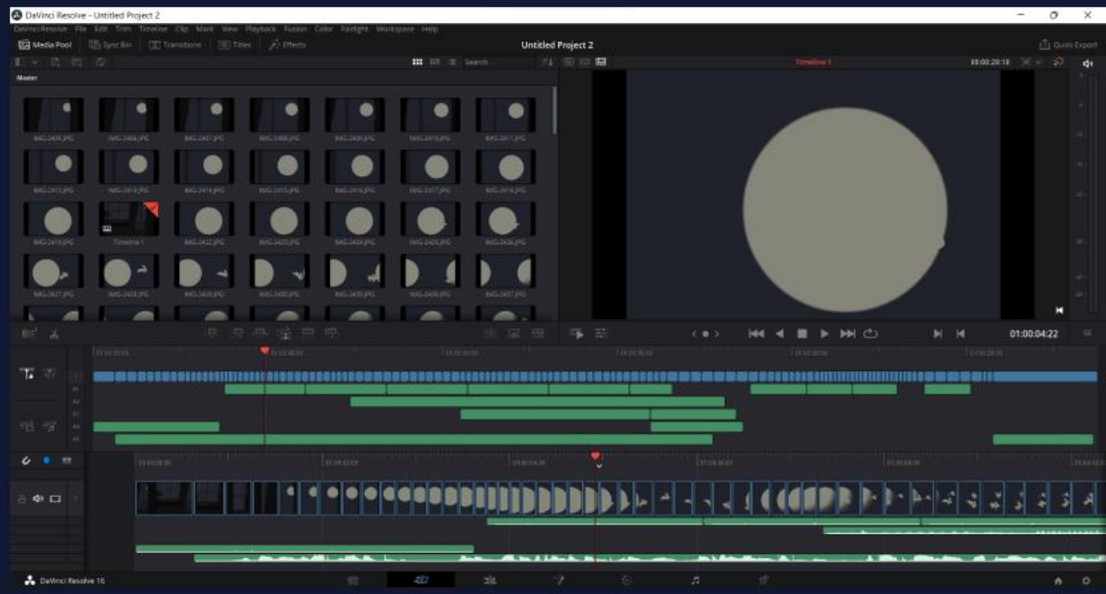
Monkeys
(from Daoshi)

Yutu
(from Hanyuefu)

Jingwei
(from The Classic of Mountains and Seas)

Xuanwu
(from Chu Ci)

Story
The animals from mythologies run across the sky every night, and before people wake up, they fall from the sky and become animal shaped biscuits.



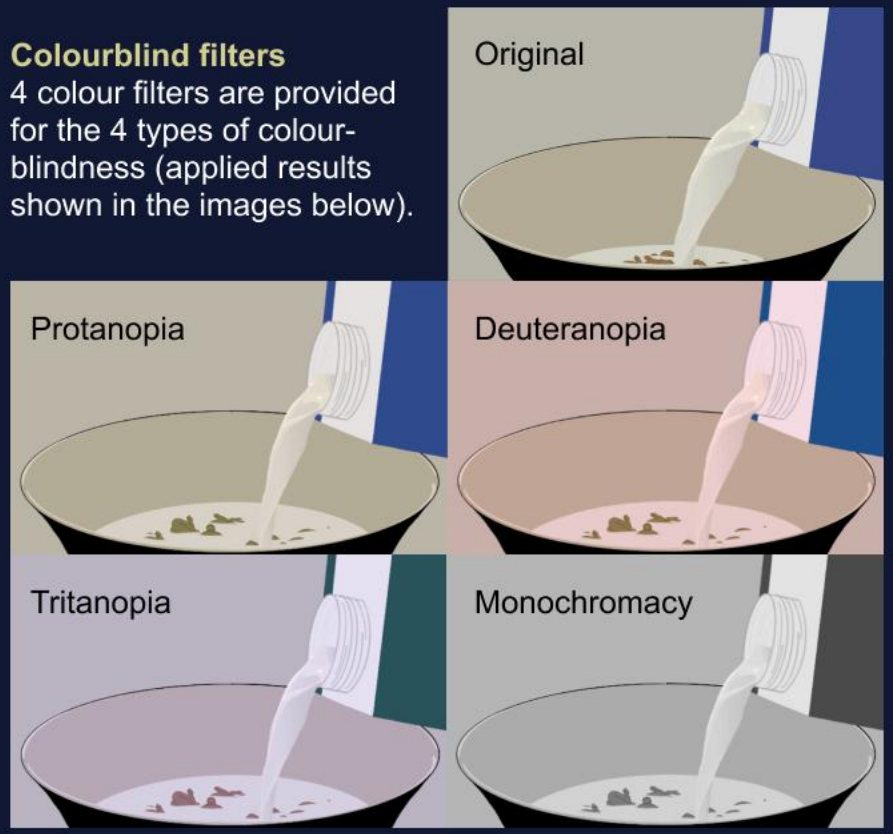
(left) Storyboard (pencil on paper)
(right) Frame-by-frame animation illustration (Procreate)

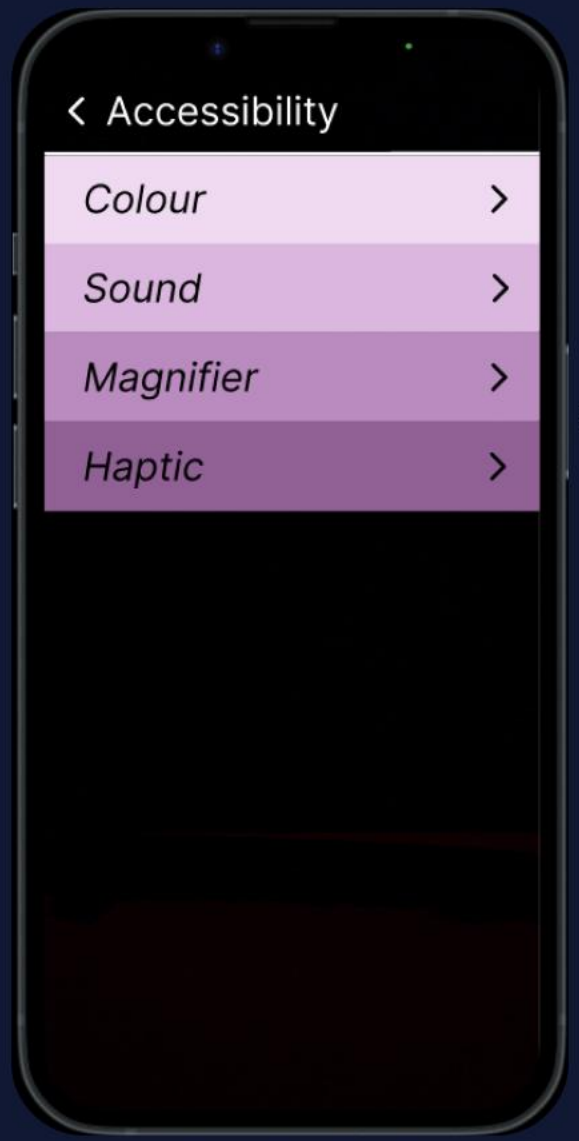
The animation was made using Da Vinci Resolve 16. Since animating on 2s is commonly used in animation studios (Lally et al, 2018), this technique is adapted here to reduce the workload.

Accessibility

Colourblind filters

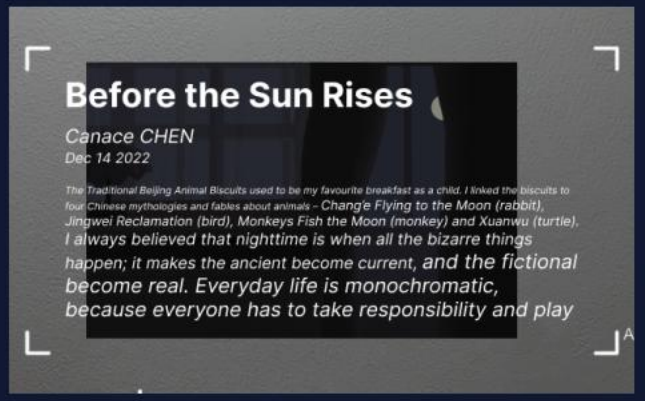
4 colour filters are provided for the 4 types of colour-blindness (applied results shown in the images below).





Dual Soundtracks

The animation provides 2 sound tracks: one with the story narrative and one with only ambient sound. The story narrative is designed to help people who needs more context in order to understand the story, especially for blind people who heavily relies on hearing when experiencing the world (Bisenzi and Carducci, 2022).



Magnifier

Enlarges any part of the interface and objects in front of the camera, making it easier for older users to see and read.

Haptic Feedback

The haptic system is enabled by default. It vibrates twice when the app is ready to scan, when the photo has been successfully recognised, when the photo is not within the frame during the animation, and when the animation is finished. It vibrates once when the user presses a button. It provides feedback to help guiding users and improve overall perception (Díaz et al, 2006).
The User can turn it off manually.

AR Interface Development

Scanning Page

Helps users align the photo print with the app's on-screen frame for accurate recognition. The app checks the surrounding lighting to ensure a successful AR experience. If the environment is too dark, users are prompted to turn on the flashlight.

Multi-sensory AR Animation

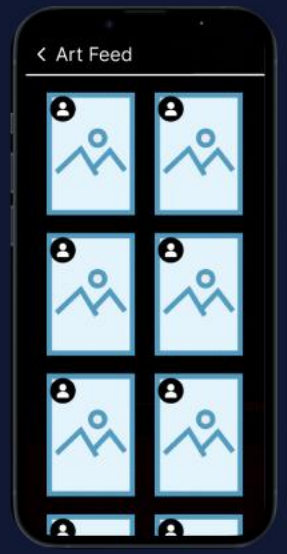
Once the photo is recognised, the Multi-sensory AR animation begins to play over the image. A red button is included for users to capture photos or record videos, creating a sharable experience.

Info

The Info page overlays essential project details on photo/animation. It displays the project name, artist(s), completion date, and the artist statement, providing additional context and background.

AR Collaboration Tool

Invites users to move their phones to engage directly with the artwork. Adding their own "graffiti" on top with the selection of brushes and colours provided. This participatory design encourages creativity, collaboration, and engagement, allowing users to become co-creators and inspire each other, all while maintaining the integrity of the original piece.



Art Feed

A curated space for inspiration. It allows users to explore other projects and artists, fostering a sense of community and promoting the exchange of creative ideas, encouraging further exploration and collaboration in the AR space.

Result Showcase